PROBLEM 20-15 A carboxylic acid has a very broad –OH absorption at 2500–3300 cm^{-1}.

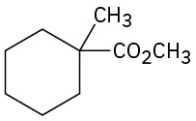
PROBLEM 20-16 4-Hydroxycyclohexanone: H–C–O absorption near 4 δ in the ^1H spectrum and C=O absorption near 210 δ in the ^{13}C spectrum. Cyclopentanecarboxylic acid: $-CO_2H$ absorption near 12 δ in the ^1H spectrum and $-CO_2H$ absorption near 170 δ in the ^{13}C spectrum.

Chapter 21

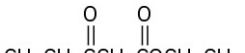
PROBLEM 21-1 **(a)** 4-Methylpentanoyl chloride **(b)** Cyclohexylacetamide **(c)** Isopropyl 2-methylpropanoate **(d)** Benzoic anhydride **(e)** Isopropyl cyclopentanecarboxylate **(f)** Cyclopentyl 2-methylpropanoate **(g)** *N*-Methyl-4-pentenamide **(h)** (*R*)-2-Hydroxypropanoyl phosphate **(i)** Ethyl 2,3-dimethyl-2-butenethioate

PROBLEM 21-2 **(a)** $C_6H_5CO_2C_6H_5$ **(b)** $CH_3CH_2CH_2CON(CH_3)CH_2CH_3$ **(c)** $(CH_3)_2CHCH_2CH(CH_3)COCl$

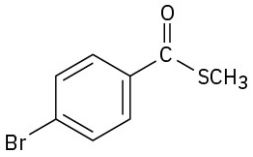
(d)



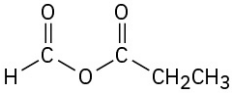
(e)



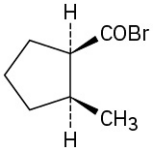
(f)



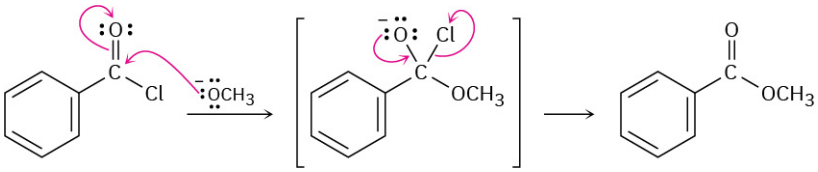
(g)



(h)



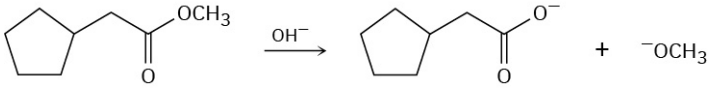
PROBLEM 21-3



PROBLEM 21-4 **(a)** Acetyl chloride > Methyl acetate > Acetamide **(b)** Hexafluoroisopropyl acetate > 2,2,2-Trichloroethyl acetate > Ethyl acetate

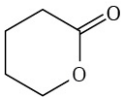
PROBLEM 21-5 **(a)** $CH_3CO_2^-\ Na^+$ **(b)** CH_3CONH_2 **(c)** $CH_3CO_2CH_3 + CH_3CO_2^-\ Na^+$ **(d)** $CH_3CONHCH_3$

PROBLEM 21-6



PROBLEM 21-7 **(a)** Acetic acid + 1-butanol **(b)** Butanoic acid + methanol **(c)** Cyclopentanecarboxylic acid + isopropyl alcohol

PROBLEM 21-8



PROBLEM 21-9 **(a)** Propanoyl chloride + methanol **(b)** Acetyl chloride + ethanol **(c)** Benzoyl chloride + ethanol

PROBLEM 21-10 Benzoyl chloride + cyclohexanol

PROBLEM 21-11 This is a typical nucleophilic acyl substitution reaction, with morpholine as the nucleophile and chloride as the leaving group.

PROBLEM 21-12 **(a)** Propanoyl chloride + methylamine **(b)** Benzoyl chloride + diethylamine **(c)** Propanoyl chloride + ammonia

PROBLEM 21-13 **(a)** Benzoyl chloride + $[(CH_3)_2CH]_2CuLi$, or 2-methylpropanoyl chloride + Ph_2CuLi **(b)** 2-Propenoyl chloride + $(CH_3CH_2CH_2)_2CuLi$, or butanoyl chloride + $(H_2C{=}CH)_2CuLi$

PROBLEM This is a typical nucleophilic acyl substitution reaction, with *p*-hydroxyaniline as the nucleophile

21-14 and acetate ion as the leaving group.

PROBLEM Monomethyl ester of benzene-1,2-dicarboxylic acid

21-15

PROBLEM Reaction of a carboxylic acid with an alkoxide ion gives the carboxylate ion.

21-16

PROBLEM $LiAlH_4$ gives $HOCH_2CH_2CH_2CH_2OH$; DIBAH gives $HOCH_2CH_2CH_2CHO$.

21-17

PROBLEM **(a)** $CH_3CH_2CH_2CH(CH_3)CH_2OH + CH_3OH$ **(b)** $PhOH + PhCH_2OH$

21-18

PROBLEM **(a)** Ethyl benzoate + 2 CH_3MgBr **(b)** Ethyl acetate + 2 $PhMgBr$

21-19 **(c)** Ethyl pentanoate + 2 CH_3CH_2MgBr

PROBLEM **(a)** H_2O, NaOH **(b)** Benzoic acid + $LiAlH_4$ **(c)** $LiAlH_4$

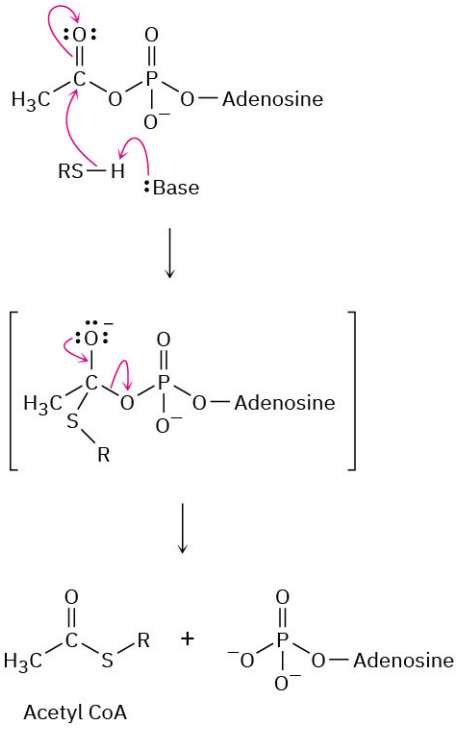
21-20

PROBLEM 1. Mg; 2. CO_2, then H_3O^+; 3. $SOCl_2$; 4. $(CH_3)_2NH$; 5. $LiAlH_4$

21-21

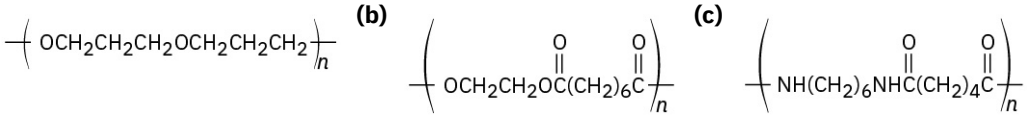
PROBLEM

21-22



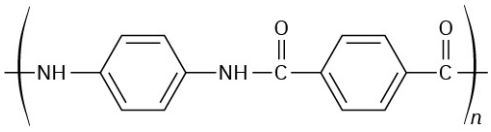
PROBLEM **(a)** **(b)** **(c)**

21-23



PROBLEM

21-24



PROBLEM **(a)** Ester **(b)** Acid chloride **(c)** Carboxylic acid **(d)** Aliphatic ketone or cyclohexanone

21-25

PROBLEM **(a)** $CH_3CH_2CH_2CO_2CH_2CH_3$ and other possibilities **(b)** $CH_3CON(CH_3)_2$

21-26 **(c)** $CH_3CH{=}CHCOCl$ or $H_2C{=}C(CH_3)COCl$